

HONOLULU PIER 4, HI, USA

WMO: 994007

Lat: 21.3033N

Lon: 157.8645W

Elev: 3

StdP: 101.29

Time Zone: -10.00 (W10)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 3	(b) 19.2	(c) 20.0	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 7.0	(k) 22.8	(l) 6.1	(m) 23.2	(n) 1.5	(o) 60	(p) 0.253

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 9	(b) 4.0	(c) 30.0	(d) N/A	(e) 29.5	(f) N/A	(g) 29.0	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 2.7	(p) 60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 5.8	(b) 5.2	(c) 4.7	(e) 17.4	(f) 30.4	(g) 0.6	(h) 0.8	(i) 17.0	(j) 31.0	(k) 16.6	(l) 31.5	(m) 16.3	(n) 31.9	(o) 15.8	(p) 32.5		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.0	23.2	23.1	23.1	24.1	25.0	25.8	26.5	26.9	26.9	26.4	25.2	24.2
	DBStd	1.73	1.02	1.26	1.20	1.11	0.95	0.71	0.89	0.86	0.88	0.85	0.99	0.95
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5488	408	366	404	424	464	475	511	523	507	508	457	440
	CDD18.3	2446	149	133	146	174	206	225	252	264	257	250	207	182
	CDH23.3	16071	406	351	346	723	1193	1681	2181	2486	2428	2161	1338	778
CDH26.7	1927	1	4	3	17	68	159	333	467	477	303	81	14	
Wind	WSAvg	2.2	2.1	2.2	2.4	2.3	2.3	2.2	2.2	2.1	2.0	2.1	2.0	
Precipitation	PrecAvg	477	69	52	51	29	26	10	14	16	21	46	60	77
	PrecMax	1086	326	240	213	227	184	62	69	194	114	283	374	439
	PrecMin	93	1	2	0	2	1	1	1	0	1	2	1	0
	PrecStd	218	74	56	47	37	34	14	15	29	20	51	73	86
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.8	27.3	27.0	27.9	28.8	29.1	30.2	30.5	30.5	30.1	29.0	27.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	26.0	26.3	26.4	27.2	28.0	28.5	29.5	29.9	30.0	29.4	28.2	27.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	25.6	25.7	25.7	26.7	27.5	28.0	28.9	29.3	29.4	28.9	27.6	26.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	25.2	25.2	25.1	26.2	26.9	27.7	28.4	28.8	28.9	28.3	27.1	26.0
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	3.9	3.9	3.8	3.8	3.9	4.0	3.9	3.9	4.0	3.9	3.7	3.8
		MCDBR	4.2	4.3	4.2	4.2	4.3	4.2	4.1	4.2	4.3	4.3	4.1	4.2
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.366	0.373	0.381	0.391	0.384	0.371	0.369	0.368	0.361	0.380	0.379	0.361
	taud		2.567	2.535	2.521	2.482	2.533	2.577	2.591	2.594	2.617	2.568	2.562	2.602
	Ebn at Noon		909	921	924	913	909	913	915	922	930	903	888	902
	Edh at Noon		93	101	107	112	106	101	99	99	96	97	93	88
All-Sky Solar Radiation	RadAvg		4.44	5.00	5.73	6.46	6.92	7.24	7.12	6.84	6.40	5.38	4.54	4.11
	RadStd		0.19	0.35	0.60	0.37	0.39	0.20	0.15	0.27	0.19	0.17	0.12	0.29

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+68	+68

Nomenclature: See separate page